

Engineering Handbook – Volume 04

API Engineering Handbook (Version 1.0)

Purpose

This document defines the REST API layer of the Energy Management SaaS platform. The API is responsible for configuration, authentication, telemetry ingestion and future industrial workflows.

Technology

FastAPI is used to expose OpenAPI-compliant REST endpoints. Automatic Swagger documentation is available through /docs. Pydantic schemas validate requests and responses.

Authentication

JWT bearer authentication protects secured endpoints. Passwords are hashed using bcrypt before storage. Future roadmap includes refresh tokens and RBAC.

Current API Modules

Current implementation contains endpoints for Health, Authentication, Companies, Users, Departments, Machines, Machine Components and Telemetry.

Endpoint Standards

Every endpoint should provide: request validation, structured JSON responses, HTTP status codes, error payloads, logging, authentication checks and OpenAPI documentation.

API Versioning

The platform should adopt versioned APIs (/api/v1/...) to preserve backward compatibility while allowing future enhancements.

Response Convention

Successful responses should return deterministic JSON objects. Error responses should include message, error code and correlation ID for troubleshooting.

Telemetry APIs

Telemetry ingestion endpoints should accept timestamped payloads from Raspberry Pi gateways, validate device ownership and store readings with minimal latency.

Future APIs

Planned APIs include Tag Definitions, Machine Types, Component Types, Device Provisioning, Alarm Management, Notification Management, Dashboards, Reports, Recipe OCR, Costing and Analytics.

Security Considerations

Rate limiting, audit logging, request tracing, tenant isolation, HTTPS-only transport, JWT expiry and API key support are recommended for production deployments.

Recommended API Structure

GET /health
POST /login
POST /users
GET /companies
POST /companies
GET /departments
POST /departments
GET /machines
POST /machines
GET /machine-components
POST /machine-components
POST /telemetry
GET /telemetry

Engineering Guidelines

- APIs must remain backward compatible.
- Never expose internal database details directly.
- Validate all incoming payloads.
- Return consistent HTTP status codes.
- Keep business logic out of routers; move complex workflows into service classes as the platform grows.
- Update Swagger and handbook whenever an endpoint changes.

Roadmap

Future editions of this volume will enumerate every endpoint directly from the repository, including request/response schemas, authentication requirements, validation rules, example payloads, sequence diagrams and implementation notes.